

MIT Vishwaprayag University, Solapur

School of Business

BS3210: Finance for Engineers, Designers and Professionals

Assignment 1: Development of a Time Value of Money (TVM) Calculator Application

Background

Financial decision-making requires understanding the Time Value of Money (TVM), one of the most fundamental concepts in finance. As part of this course, students will design and develop a functional application capable of performing Time Value of Money calculations for real-life financial decisions.

This assignment serves as the first milestone toward developing a comprehensive Financial Services Assistance Application.

Assignment Objective

Develop a working application that enables users to perform the following financial calculations:

1. Future Value (FV)
2. Present Value (PV)
3. Simple Interest
4. Compound Interest
5. EMI Calculation
6. Loan Amortization Schedule
7. Scenario Analysis based on different interest rates

The application will demonstrate financial accuracy by application of software development skills.

Permitted Submission Formats ONLY

Students must submit the project as an HTML-Based Web Application

Only executable HTML/Web Applications will be evaluated.

Functional Requirements

The application must provide the following Calculators.

Calculator 1: Future Value Calculator

Inputs

- Principal Amount
- Annual Interest Rate

- Time Period (Years)

Output

- Future Value

Calculator 2: Present Value Calculator

Inputs

- Future Value
- Discount Rate
- Time Period

Output

- Present Value

Calculator 3: Simple Interest Calculator

Inputs

- Principal
- Interest Rate
- Time

Output

- Interest Earned
- Maturity Value

Calculator 4: Compound Interest Calculator

Inputs

- Principal
- Rate
- Time
- Compounding Frequency

Output

- Compound Interest
- Maturity Value

Calculator 5: EMI Calculator

Inputs

- Loan Amount
- Interest Rate
- Loan Tenure

Output

- EMI
- Total Payment
- Total Interest Payable

Calculator 6: Loan Amortization Schedule

Inputs

- Loan Amount
- Interest Rate
- Tenure

Output Table

Month	Opening Balance	Interest	Principal Repaid	EMI	Closing Balance
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The schedule must be generated automatically.

Calculator 7: Scenario Analysis

The application should allow users to compare outcomes under at least four interest rates:

- 8%
- 10%
- 12%
- 15%

Submission Requirements:**Classroom Demonstration and Evaluation**

All teams will demonstrate their application during the scheduled evaluation session.

Marks will be awarded based on live testing performed by the faculty.

Students must ensure their application is fully functional on the day of the demonstration.

Evaluation Rubric (20 Marks) (Converted to 5 Marks of CIA)

Criterion	Marks
Functional Accuracy of Calculations	5
Correct Financial Formula Implementation	5
User Interface and User Experience	4
Input Validation and Error Handling	3
Performance During Live Testing	3
Total	20

Important Note

This assignment focuses equally on:

1. Financial understanding,
2. Problem-solving ability,
3. Software development skills, and
4. User-centric application design.

Students are encouraged to create professional-quality applications suitable for real-world financial decision-making.